

20260224EB_MN (NCT, ATP, TDP43, IF, RNA)

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- 2026年2月24日 星期二
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2026年2月26日 星期四
- T2 -> T3
- 2026年2月28日 星期六
- T3 -> T4
- 2026年3月2日 星期一
- T4 -> T5
- 2026年3月4日 星期三
- T5 -> T6
- 2026年3月6日 星期五
- T6 -> T6 (~14mL for over-weekend)
Coat PLO
- 2026年3月7日 星期六
- Coat laminin
- 2026年3月9日 星期一
- T6 -> T7
Dissociation (process dishA and dishB separately)
 1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
 2. Remove sup. and add 4mL 0.25% trypsin (prepare 8mL 0.25% trypsin with 100U DNaseI)
 3. Pipette ~8-10 with P1000 and then incubate the tubes in 37C.
 4. pipette*10 at 5min, 10min, 15min, and 20min
 5. add 8mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times
 6. add 8mL CTWM -> pipette ~ 10 times

7. filter cells through 40um cell strainer into new tubes.
8. centrifuge at 200xg, 4min

cell info				^
	A	B	C	
1	cell	passage	well#	
2	WT11	P4	5	

Table1			^
	A	B	
1		volume (mL)	
2	trypsin	4	
3	HI-FBS/CTWM	8	

Table3											/
	A	B	C	D	E	F	G	H	I	J	
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume	
2	dish A	1.35	89	5	6.75	I	24	225	24		
3	dish B	1.25	90	5	6.25	II	24	225	6		
4						III	8-well chamber slide	100	8		
5						IV	6 (A1-3)	1375	2		
6							6 (B1-3)	1350	2		
7				cell#	well#	volume					
8	vLH92	A	plate I	225	5	833	with 6.25uL vLH92				
9		B	plateI and II	225	11	1980	with 13.8uL vLH92				
10	no virus	A	plate I	225 100	3 4	796					
11		B	plate I	225 100	3 4	860					
12	110&111	A	plate I	225	4	667	with 12uL vLH110 with 18uL vLH111				

Plate I ^

	1	2	3	4	5	6
A	A, 110 (12uL total)		A (92)	A	A	A
B			A	A (no virus)		
C	A, 111 (18uL total)		B (92)	B (no virus)		
D			B	B	B	B

Plate II (from last time, for Theo) ^

	1	2	3	4	5	6
A						
B						
C				B (92)		
D						

Plate III (8-well chamber slide with 1.5 bottom) ^

	A	B	C	D
1	A		B	
2				

Plate IV ^

	1	2	3
A	A	A	
B	B	B	

2026年3月11日 星期三

Change T7 -> T7 if added virus

2026年3月12日 星期四

T7-> T7 if not added virus

2026年3月13日 星期五

T7 -> T8

2026年3月14日 星期六

1/2 T8 -> 1/2 T8

2026年3月17日 星期二

1/2 T8 -> 1/2 T8

☒ Treat cells for 24h timepoint from 11:40 - 11:45

2026年3月18日 星期三

Image NCT (1h and 24h)

1. Aliquote 850 uL and 820 uL into 1.5mL eppendorf.
2. Add SirDNA (1:5000) with treatment (0.32 uL into 6.4 uL media. Add 1.6uL into 400uL media on the plate for 24h treatment. Add 3.2 uL into 820uL media for 1h treatment)

NCT_O.C. , 60X, 600x900, 16-bit						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	561	16	30	300 ms	the signal is good	
3	640	16	15	100 ms		
4	488	16	50	1s		

time table for NCT treatment																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
1	condition#		condition	media	time	imaging												
2	on 3/18/2026		110															
3	1	A1	2.5mM glucose 1h	2.5mM gluc	8:05	9:05												
4	2	A2	0.15mM glucose 1h	0.15mM gluc	8:35	9:35												
5	3	B1	2.5mM glucose 24h	2.5mM gluc	3/17 11:40	11:04												
6	4	B2	0.15mM glucose 24h	0.15mM gluc	3/17 11:40	11:36												
7			111															
8	1	C 1	2.5mM glucose 1h	2.5mM gluc	9:05	10:08												
9	2	C 2	0.15mM glucose 1h	0.15mM gluc	9:35	10:35												
10	3	D 1	2.5mM glucose 24h	2.5mM gluc	3/17 11:45	12:05												
11	4	D 2	0.15mM glucose 24h	0.15mM gluc	3/17 11:45	12:30												

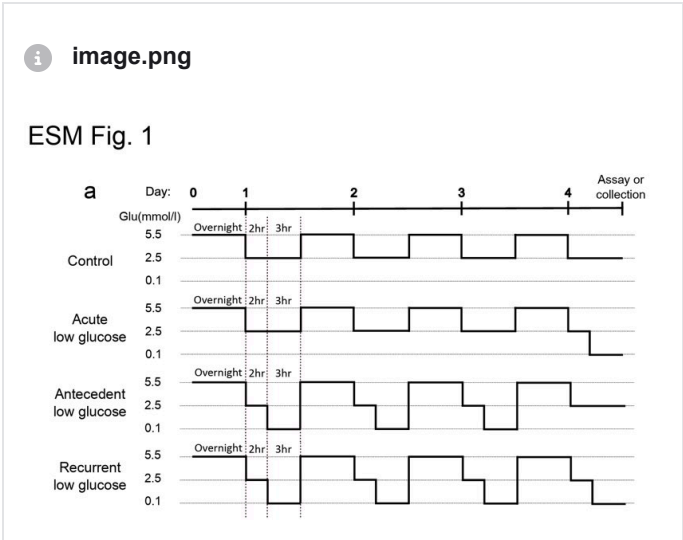
fileID_treatment_NCT						
	A	B	C	D	E	F
1	image	reporter	treatment	time	type	note
2	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	2.5mM gluc	1h	no-change	
3	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	2.5mM gluc	1h	no-change	
4	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0.15mM gluc	1h	no-change	
5	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0.15mM gluc	1h	no-change	
6	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	2.5mM gluc	24h	no-change	
7	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	2.5mM gluc	24h	no-change	
8	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	0.15mM gluc	24h	no-change	
9	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	0.15mM gluc	24h	no-change	
10	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	2.5mM gluc	1h	no-change	
11	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	2.5mM gluc	1h	no-change	
12	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0.15mM gluc	1h	no-change	
13	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0.15mM gluc	1h	no-change	
14	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	2.5mM gluc	24h	no-change	
15	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	2.5mM gluc	24h	no-change	
16	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	0.15mM gluc	24h	no-change	
17	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	0.15mM gluc	24h	no-change	

RNA with Recurrent Low Glucose (RLG)

☒ D1 treatment

Long-term recurrent test for RNA

ref: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6290858/#Abs1>



Use the first and forth row (Control vs. RLG)

Prepare 250mL no-gluc Neurobasal-A media with N2, B27 w/o vitA, Glutamax

For each 50mL
250mL no-gluc NBA media
2.5 mL N2
5 mL B27 w/o vitA
2.5 mL GlutaMAX
2.5mL P/S
--> filtered

Media needed for RLG
2.5mM gluc: 6*4(day)*2mL = 48mL
0.15mM gluc: 2*4(day)*2mL = 16mL
5.5mM gluc: 4*4(day)*2mL = 32mL

Prepare 50mL media for each

2.5mM gluc: 50mL + 113.64uL 1.1M glucose (1:440)
0.15mM gluc: 50mL + 6.82uL 1.1M glucose (1:7333)
5.5mM gluc: 50mL + 250uL 1.1M glucose (1:200)

treatment for Plate IV			
	1	2	3
A	Control	RLG	
B	Control	RLG	

time table for RLG protocol																				
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	condition#		condition	3/18	2.5mM 2h		3h treatment	5.5mM gluc o/n	3/19	2.5mM 2h		3h treatment	5.5mM gluc o/n	3/20	2.5mM 2h		3h treatment	5.5mM gluc o/n	3/21	2.5mM 2h
2	6-well plate			day1	0h				day2	0h				day3	0h				day4	0h
3		A1	Control	2.5mM gluc	12:20	2.5mM gluc	14:20	17:20	2.5mM gluc	12:00	2.5mM gluc	14:10	18:15	2.5mM gluc	12:44	2.5mM gluc	14:43		2.5mM gluc	
4		A2	RLG	2.5mM gluc	12:20	0.15mM gluc	14:20	17:20	2.5mM gluc	12:00	0.15mM gluc	14:11	18:16	2.5mM gluc	12:44	0.15mM gluc	14:43		2.5mM gluc	2.5mM gluc
5		B1	Control	2.5mM gluc	12:20	2.5mM gluc	14:20	17:20	2.5mM gluc	12:00	2.5mM gluc	14:12	18:17	2.5mM gluc	12:44	2.5mM gluc	14:43		2.5mM gluc	0.15mM gluc
6		B2	RLG	2.5mM gluc	12:20	0.15mM gluc	14:20	17:20	2.5mM gluc	12:00	0.15mM gluc	14:13	18:18	2.5mM gluc	12:44	0.15mM gluc	14:43	17:45-48	2.5mM gluc	0.15mM gluc
																			12:30	18:00

2026年3月19日 星期四

- 1/2 T8 -> 1/2 T8
- ☒ RLG D2 treatment
- ☒ I accidentally treated hypoglycemia for 4 hours instead of 3 hours before I switched to 5.5mM gluc

IF

Table2

	A	B	C	D
1	2.5mM	2.5mM	2.5mM	2.5mM
2	0.15mM	0.15mM	0.15mM	0.15mM

1. Treat cells as above at 14:35-37
2. Fix cells at 15:37
3. 4% PFA for 15min at RT (1200uL with 150uL 32% PFA)
4. 0.2% Triton for 10min
5. 10%FBS blocking for 1h
6. Primary antibody

Table7

	A	B	C	D	E	F	G	H	I	J	K	L
1	condition	Ab#	antibody	species	dilution	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	33	TDP-43	Rb	500	400	0.8	488	B	1000	800	0.8
3	A	147	Importin-b1	m	1000	400	0.4	594	O	1000	800	0.8
4	B	C-31	RANGAP1	Rb	500	800	1.6	488	B			
5	B	146	Ran	m	500	800	1.6	594	O			
6	C	82	HNRNPA2B1	Rb	500	400	0.8	594	B			
7	C	147	Importin-b1	m	1000	400	0.4	594	O			

Table4

	A
1	SC-35 Nuclear Speckle Marker
2	YAP Antibody (63.7)

2026年3月20日 星期五

Imaging TDP43 (9h recurrent hypoglycemia)

Prepare media

1. I treated SPY505 (1:5000) together with hypoglycemia and JFX650 (1nM)
- a. 0.5 uL SPY505 + 9.5uL 0.15mM gluc media

- b. Add 3.2 into 800uL media
- c. I accidentally used all 2.5mM + SPY505, so I prepared 800uL 2.5mM with 0.16uL SPY505 again (but I think there is pipetting error with such small amount)

2. JFX650

- a. 2.5mM: 3.2mL +1.28uL 2.5uM JFX650
- b. 0.15mM: 6.4mL +2.56uL 2.5uM JFX650

Well1						
	1	2	3	4	5	6
A						
B				1	2	3
C				1	2	3
D						

20260320_timetable													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1		B4		B5		B6		C4		C5		C6	
2	9:00	treat SPY505		treat SPY505		treat SPY505							
3													
4	10:00	treat#1	2.5mM +JFX	treat#1	0.15mM +JFX	treat#1	0.15mM +JFX						
5	10:30							treat#1	2.5mM +JFX	treat#1	0.15mM +JFX	treat#1	0.15mM +JFX
6	11:00	image@1h		image@1h		image@1h							
7	11:30	treat#2	2.5mM gluc	treat#2	0.15mM gluc	treat#2	2.5mM gluc	image@1h		image@1h		image@1h	
8	12:00							treat#2	2.5mM gluc	treat#2	0.15mM gluc	treat#2	2.5mM gluc
9	12:30	treat 6-well											
10	13:00												
11	13:30												
12	14:00	treat#3	2.5mM gluc	treat#3	0.15mM gluc	treat#3	0.15mM gluc						
13	14:30	treat 6-well						treat#3	2.5mM gluc	treat#3	0.15mM gluc	treat#3	0.15mM gluc
14	15:00	treat#4	2.5mM gluc	treat#2	0.15mM gluc	treat#4	2.5mM gluc						
15	15:30							treat#4	2.5mM gluc	treat#2	0.15mM gluc	treat#4	2.5mM gluc
16	16:00												
17	16:30												
18	17:00												
19	17:30	treat 6-well											
20	18:00	treat#5	2.5mM +JFX	treat#5	0.15mM +JFX	treat#5	0.15mM +JFX						
21	18:30							treat#5	2.5mM +JFX	treat#5	0.15mM +JFX	treat#5	0.15mM +JFX
22	19:00	image@9h		image@9h		image@9h							
23	19:30							image@9h		image@9h		image@9h	

Table5																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
1				treat#1			treat#2		treat#3		treat#4		treat#5					
2				0h		1h			4h		5h		8h		9h		24h	
3	condition#		condition	media+JFX65 0+SPY505	time	imaging	media after imaging	time	media	time	media	time	media	time	imaging	JFX650	imaging on 2/28	
4	batch A																	
5		1	B4	2.5mM glucose constant	2.5mM gluc	10:05	11:10 (take the image twice)	2.5mM gluc	11:15	2.5mM gluc	14:00	2.5mM gluc	15:00	2.5mM gluc	18:00	19:08	10:08	11:12
6		2	B5	0.15mM glucose recurrent	0.15mM gluc	10:20	11:27	2.5mM gluc	11:29	0.15mM gluc	14:17	2.5mM gluc	15:17	0.15mM gluc	18:15	19:20	10:22	11:46
7		3	B6	0.15mM glucose constant	0.15mM gluc	10:35	11:40	0.15mM gluc	11:43	0.15mM gluc	14:30 (treat 6- well soon after)	0.15mM gluc	15:33	0.15mM gluc	18:35	19:37	10:43	11:57
8	batch B																	
9		1	C4	2.5mM glucose constant	2.5mM gluc	10:45	11:54	2.5mM gluc	11:58	2.5mM gluc	14:45	2.5mM gluc	15:45	2.5mM gluc	18:50	19:52	10:57	12:11
10		2	C5	0.15mM glucose recurrent	0.15mM gluc	11:12	12:13	2.5mM gluc	12:15	0.15mM gluc	15:10	2.5mM gluc	16:10	0.15mM gluc	19:10	20:24	11:20	12:26
11		3	C6	0.15mM glucose constant	0.15mM gluc	11:32	12:32	0.15mM gluc	12:35 (treat 6- well after this)	0.15mM gluc	15:18	0.15mM gluc	16:20	0.15mM gluc	19:25	20:37	11:48	12:51
12																		

TDP43_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	488	16	20	300 ms	The SPY505 signal was pretty bad for 1h (except C4 since I prepared it separately, and I might pipette a bit more SPY505 into it.)
3	640	16	10	300 ms	

fileid_treat_TDP43

	A	B	C	D	E
1	image	treatment	treat_time	type	replicate
2	MN_TDP43Halo_60x_0001	2.5mM gluc	1h	constant	A
3	MN_TDP43Halo_60x_0002	0.15mM gluc	1h	recurrent	A
4	MN_TDP43Halo_60x_0003	0.15mM gluc	1h	constant	A
5	MN_TDP43Halo_60x_0004	2.5mM gluc	1h	constant	B
6	MN_TDP43Halo_60x_0005	0.15mM gluc	1h	recurrent	B
7	MN_TDP43Halo_60x_0006	0.15mM gluc	1h	constant	B
8	MN_TDP43Halo_60x_0007	2.5mM gluc	9h	constant	A
9	MN_TDP43Halo_60x_0008	0.15mM gluc	9h	recurrent	A
10	MN_TDP43Halo_60x_0009	0.15mM gluc	9h	constant	A
11	MN_TDP43Halo_60x_0010	2.5mM gluc	9h	constant	B
12	MN_TDP43Halo_60x_0011	0.15mM gluc	9h	recurrent	B
13	MN_TDP43Halo_60x_0012	0.15mM gluc	9h	constant	B
14	MN_TDP43Halo_60x_0013	2.5mM gluc	24h	constant	A
15	MN_TDP43Halo_60x_0014	0.15mM gluc	24h	recurrent	A
16	MN_TDP43Halo_60x_0015	0.15mM gluc	24h	constant	A
17	MN_TDP43Halo_60x_0016	2.5mM gluc	24h	constant	B
18	MN_TDP43Halo_60x_0017	0.15mM gluc	24h	recurrent	B
19	MN_TDP43Halo_60x_0018	0.15mM gluc	24h	constant	B

RLG

☒ D3

2026年3月21日 星期六

ATP/ADP Imaging

Well2

	1	2	3	4	5	6
A			1	2	3	4
B			5			
C			5			
D			1	2	3	4

Experiment options:

1. [gluc] vs. ATP/ADP with pyruvate (just repeat some concentration as 20260217MN)
2. [gluc] vs. ATP/ADP in no-pyruvate context (1h)
3. low glucose w/ or w/o pyruvate +/- oligomycin time course (to test if the remaining ATP is due to the mitochondrial respiration buffer from pyruvate)
4. normal glucose or low glucose +/- oligomycin (check how the cells reply on mit with different glucose level)

time table for treatment 2026/3/1											
	A	B	C	D	E	F	G	H	I	J	K
1			batch A					batch B			
2	condition#		treatment	treat	image			treatment	treat	image	
3	1	A3	2.5mM gluc with pyruvate 1h	15:05	16:14		D3	2.5mM gluc with pyruvate 1h	16:15	17:15	
4	2	A4	0.15mM gluc with pyruvate 1h	15:15	16:29		D4	0.15mM gluc with pyruvate 1h	16:45	17:46	* attention the order here
5	3	A5	2.5mM gluc without pyruvate 1h	15:30	16:41		D5	2.5mM gluc without pyruvate 1h	16:30	17:30	* attention the order here
6	4	A6	0.15mM gluc without pyruvate 1h	15:45	16:52		D6	0.15mM gluc without pyruvate 1h	16:56	18:01	
7	5	B3	0mM gluc without pyruvate 1h	16:00	17:05		C4	0mM gluc without pyruvate 1h	17:17	18:26	

prepare media with different conc. of glucose1							
	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	1	1100	2.5	440	2200	5.0	2195.0
3	2	2.5	0.15	16.6666666667	2000	120.0	1880.0
4	3		0		2000		

fileid_treat_ATP					
	A	B	C	D	E
1	image	treatment	treat_time	pyruvate	replicate
2	vLH92_MN_60x-1xzoom_0001	2.5mM gluc	1h	yes	A
3	vLH92_MN_60x-1xzoom_0002	0.15mM gluc	1h	yes	A
4	vLH92_MN_60x-1xzoom_0003	2.5mM gluc	1h	no	A
5	vLH92_MN_60x-1xzoom_0004	0.15mM gluc	1h	no	A
6	vLH92_MN_60x-1xzoom_0005	0mM gluc	1h	no	A
7	vLH92_MN_60x-1xzoom_0006	2.5mM gluc	1h	yes	B
8	vLH92_MN_60x-1xzoom_0007	2.5mM gluc	1h	no	B
9	vLH92_MN_60x-1xzoom_0008	0.15mM gluc	1h	yes	B
10	vLH92_MN_60x-1xzoom_0009	0.15mM gluc	1h	no	B
11	vLH92_MN_60x-1xzoom_0010	0mM gluc	1h	no	B

I fixed the cells (B4-6 & C4-6) around 19:00 after I finished ATP/ADP imaging on 3/21/2026 --> to check TDP43 IF

1. 4%PFA, 15min at RT
2. Wash with PBS once
3. 0.2%Triton for 10min at RT
4. Blocking with 10% FBS in cold room

2026年3月24日 星期二

IF

Well3						
	1	2	3	4	5	6
A						
B				A		
C						
D						

Table6														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	condition	Ab#	antibody	species	dilution	volume/well	well#	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	33	TDP-43	Rb	500	166	6	1000	2	568	C	1000	600	0.6

- 2026年3月25日 星期三
1. PBS wash*3

2. 2nd Ab (C: 1:1000) for 1h at RT

3. PBS wash *3

2026年4月8日 星期三

IF imaging

Most well in 8-well chamber detached. Also, there are aa lot of bacteria contamination. --> do not image

Only image 24-well plate TDP43 staining. Also have bacteria contamination, but the cell conditions (still attached, but still A LOT of contamination) are better than 8-well chambers. Only image B4-6. Mount all of them after imaging.

IF_O.C. , 60X, 600x900, 16-bit (old confocal)						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	488	16	30	300 ms		
3	561	16	30	300 ms		
4	640	16	30	300 ms		